



IDE & Toolkit Configuration

Criterion 5 · Phase 2: Environment & GitHub · 6 Marks

SkillsTrack Learner Support Portal — JavaScript Programmer (NSSA TPC)



PROBLEMS 1

```
ESLint: Identifier already  
declared [Ln 2, Col 5]
```

Evidence walkthrough: Extensions · ESLint setup · Format on save · Linter catch · Formatter · Breakpoint · Framework

What This Criterion Wants

Proof your dev environment actually works — not just that it's installed.



Extensions installed

ESLint and Prettier added to VS Code and active.



Linters catching an error

A real mistake underlined, with the Problems panel showing it.



Formatter before/after

Messy code, then clean — reformatted automatically on save.



Breakpoint working

Execution paused mid-run with the Variables panel visible.



Framework/library installed

package.json or an import line proving a real dependency.



5 screenshots minimum

Every one shows the tool doing its job, not just sitting installed.

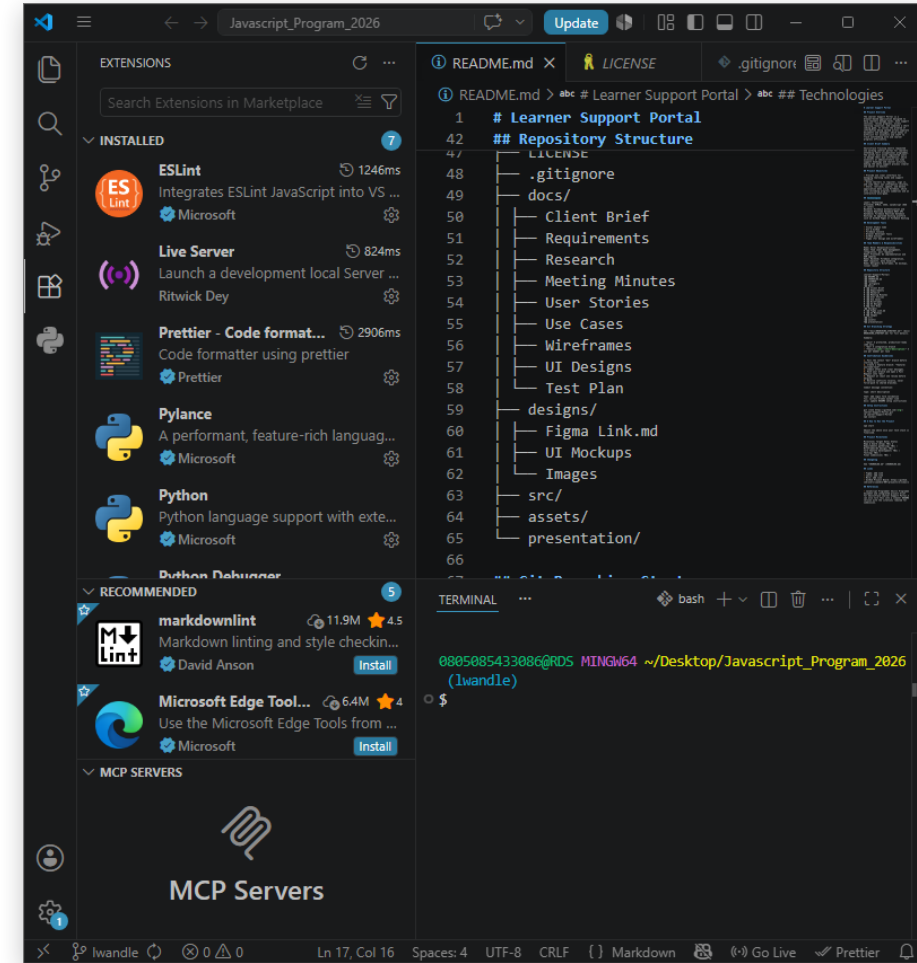
Extensions Installed

Setting Up the Toolkit



VS Code, equipped with the tools the rubric asks for.

- ESLint (by Microsoft) — catches syntax and logic issues live
- Prettier — Code formatter — auto-formats on save
- Both show as Installed in the Extensions panel
- Repository structure also visible — docs, designs, src all organised



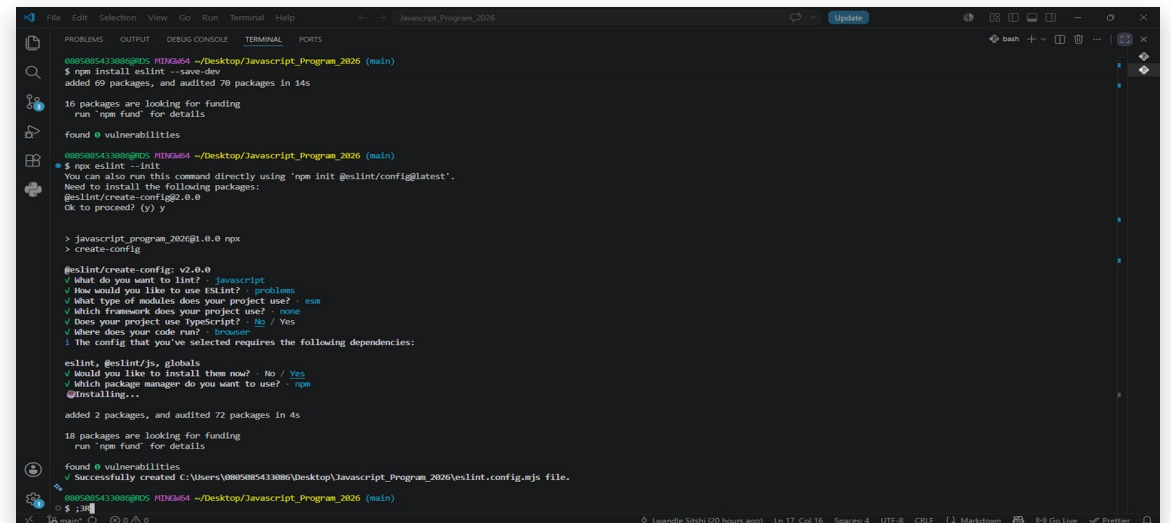
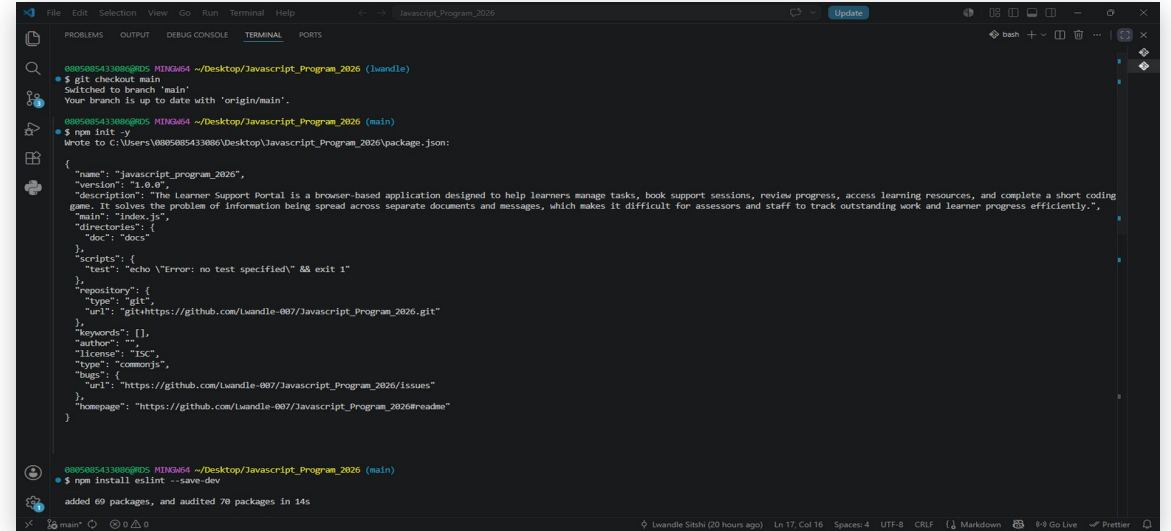
ESLint, Configured

Terminal Setup



The linter isn't just installed — it's wired into the project.

- `npm init -y` creates `package.json` for the project
- `npm install eslint --save-dev` adds ESLint as a dev dependency
- `npx eslint --init` walks through the setup prompts
- Environment: browser · Modules: JavaScript (ESM) · No TypeScript



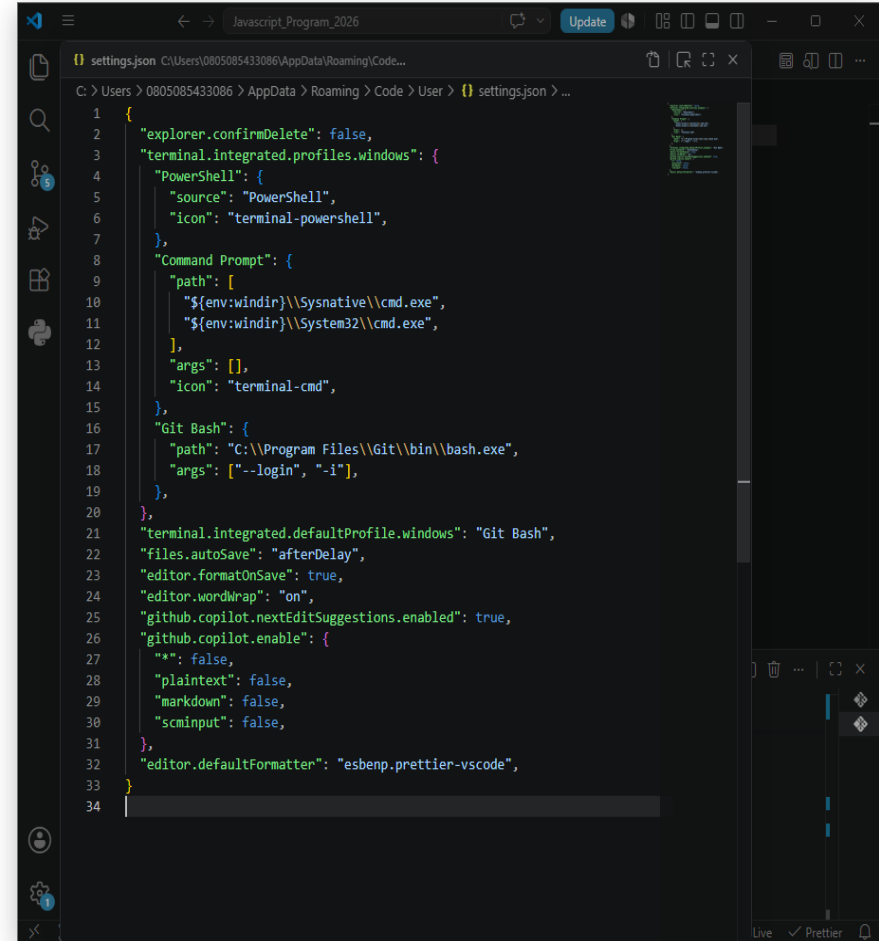
Format on Save

Editor Settings



Prettier is set as the default formatter, running automatically.

- settings.json sets "editor.formatOnSave": true
- "editor.defaultFormatter": "esbenp.prettier-vscode" points at Prettier
- Every save now reformats the file — no manual step needed
- Other quality-of-life settings configured too (word wrap, autosave)



```
1 {
2   "explorer.confirmDelete": false,
3   "terminal.integrated.profiles.windows": {
4     "PowerShell": {
5       "source": "PowerShell",
6       "icon": "terminal-powershell",
7     },
8     "Command Prompt": {
9       "path": [
10        "${env:windir}\\Sysnative\\cmd.exe",
11        "${env:windir}\\System32\\cmd.exe",
12      ],
13      "args": [],
14      "icon": "terminal-cmd",
15    },
16     "Git Bash": {
17       "path": "C:\\Program Files\\Git\\bin\\bash.exe",
18       "args": ["--login", "-i"],
19     },
20   },
21   "terminal.integrated.defaultProfile.windows": "Git Bash",
22   "files.autoSave": "afterDelay",
23   "editor.formatOnSave": true,
24   "editor.wordWrap": "on",
25   "github.copilot.nextEditSuggestions.enabled": true,
26   "github.copilot.enable": {
27     "**": false,
28     "plaintext": false,
29     "markdown": false,
30     "scminput": false,
31   },
32   "editor.defaultFormatter": "esbenp.prettier-vscode",
33 }
34
```

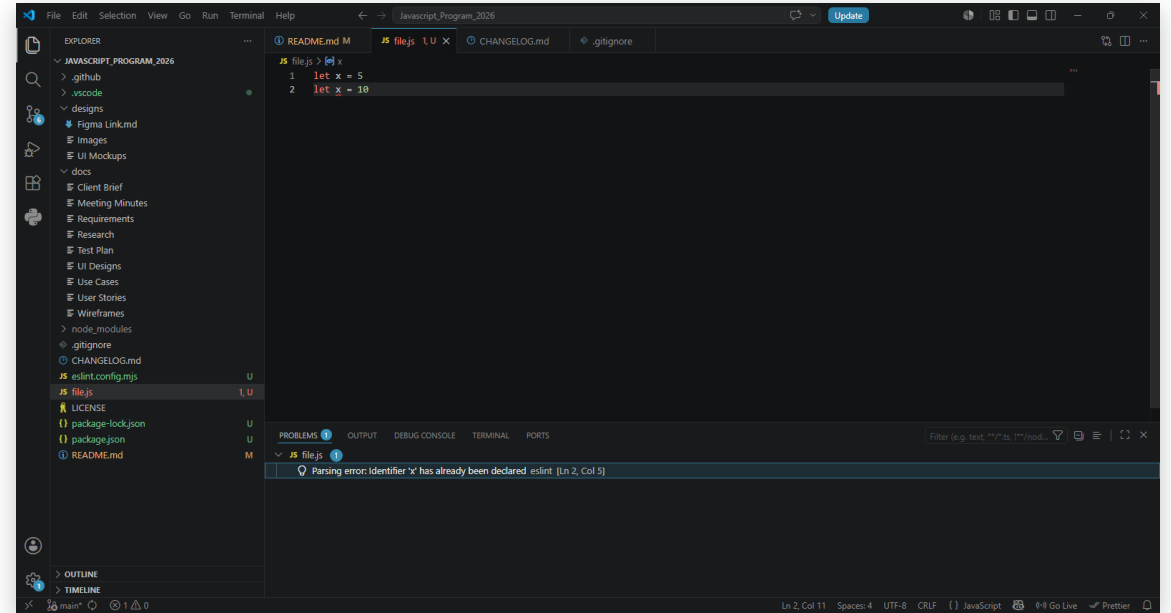
Linter, Catching an Error

Live Error Detection



A real mistake, flagged the moment it's typed.

- `let x = 5` followed by `let x = 10` redeclares the same identifier
- A red squiggly line appears under the second declaration
- The Problems panel spells it out: “Identifier 'x' has already been declared”
- Exact location given — Ln 2, Col 5 — tied to the eslint rule



file.js — duplicate declaration, ESLint parsing error shown in the Problems tab.

Formatter, Before & After

Auto-Reformatting



Deliberately messy code, cleaned up on save — automatically.

- Before: `function add(a,b){return a+b}` — no spacing, no structure
- Save triggers Prettier via format-on-save
- After: proper spacing, braces on their own lines, consistent style
- No manual cleanup — the formatter did all the work

BEFORE

```
JS file.js > add
1 function add(a,b){return a+b}
```

AFTER

```
JS file.js > ...
1 function add(a, b) {
2   return a + b;
3 }
4
```

Formatter ran on save — no manual editing between the two screenshots.

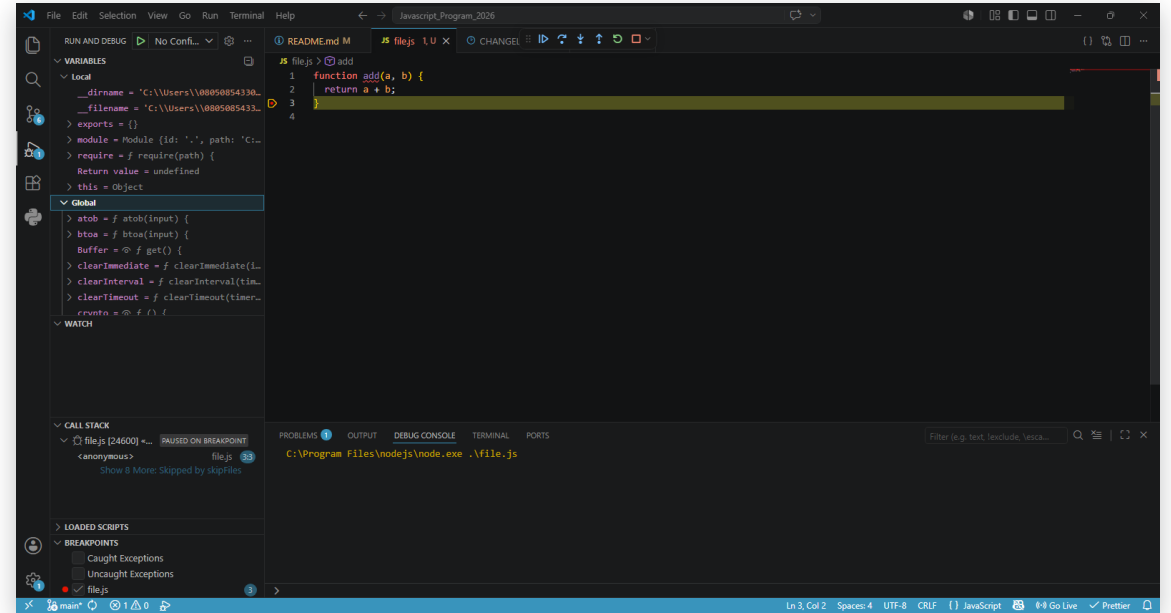
Breakpoint, Working

Debugger in Action



Execution actually pauses — and the state is inspectable.

- A breakpoint set on the closing brace of `add(a, b)`
- F5 launches the Node.js debugger
- Execution pauses exactly there — “PAUSED ON BREAKPOINT” shown
- Variables and Call Stack panels confirm the live program state



Debugger paused on `file.js`, Variables panel open on the left as required.

Framework/Library Installed

Real Dependencies



package.json proves the toolkit is more than a blank folder.

- devDependencies lists eslint, @eslint/js and globals
- Versions pinned — reproducible installs for every teammate
- repository, bugs and homepage fields point at the real GitHub repo
- Proof the toolkit is installed, tracked, and actually used

```
1 {
2   "name": "javascript_program_2026",
3   "version": "1.0.0",
4   "description": "The Learner Support Portal is a browser-based application designed to help learners manage tasks, book support sessions, review progress, access learning resources, and complete a short coding game. It solves the problem of information being spread across separate documents and messages, which makes it difficult for assessors and staff to track outstanding work and learner progress efficiently.",
5   "main": "index.js",
6   "directories": {
7     "doc": "docs"
8   },
9   "scripts": {
10     "test": "echo \\\"Error: no test specified\\\" && exit 1"
11   },
12   "repository": {
13     "type": "git",
14     "url": "git+https://github.com/Lwandle-007/Javascript_Program_2026.git"
15   },
16   "keywords": [],
17   "author": "",
18   "license": "ISC",
19   "type": "commonjs",
20   "bugs": {
21     "url": "https://github.com/Lwandle-007/Javascript_Program_2026/issues"
22   },
23   "homepage": "https://github.com/Lwandle-007/Javascript_Program_2026#readme",
24   "devDependencies": {
25     "@eslint/js": "^10.0.1",
26     "eslint": "^10.8.1",
27     "globals": "^17.10.0"
28   }
29 }
```

package.json — devDependencies confirm ESLint is a tracked project dependency.

Criterion 5 — Covered

Five working screenshots, not five installs.

- ✓ ESLint and Prettier extensions installed and active
- ✓ ESLint configured via terminal — `npm init`, `install`, `eslint --init`
- ✓ Format-on-save wired to Prettier in `settings.json`
- ✓ Linter caught a real duplicate-declaration error
- ✓ Formatter cleaned messy code automatically on save
- ✓ Breakpoint paused execution with Variables panel visible
- ✓ `package.json` lists ESLint as a real, tracked dependency

6

out of 6 marks

*"Screenshots proving these tools were only installed
are not enough — every one must show the tool doing
its job."*